KHLDGNet: Knowledge-Driven Hierarchical Language-Aware Domain Generalization Network for Cross-Scene Wetland Hyperspectral Image Classification

Yining Feng, Zhenhua Mu, Yin Zheng, Chuanming Song, and Xianghai Wang

***Abstract*—Wetlands are fragile yet vital ecosystems that require precise remote sensing (RS) mapping. Hyperspectral image (HSI) provides high spectral resolution for fine-scale wetland classification, but cross-scene application suffers from domain shifts and a lack of model interpretability. To address these issues, this article proposes a knowledge-driven hierarchical language-aware domain generalization network (KHLDGNet) for interpretable cross-scene wetland HSI classification. Unlike existing vision–language methods that merely use category names or simple attributes as auxiliary text, the core innovation of our framework is the construction of a structured, four-level geographic knowledge hierarchy that encompasses conceptual, rule, instance, and causal. This hierarchy transforms a geographical cognitive framework into computable semantic constraints, guiding the model from superficial pattern matching to deep mechanistic understanding. A MixStyle module in the visual encoder implicitly diversifies feature-level styles to counter spectral and textural variations. Visual and textual features are aligned in a shared semantic space via supervised contrastive learning, unifying generalization and interpretability. Experiments on three real-world wetland datasets demonstrate that KHLDGNet significantly outperforms state-of-the-art domain generalization (DG) methods. Visualization and case studies confirm its semantic explanation capability, contributing to explainable geographic AI (XGeoAI). Code available at https://github.com/SYFYN0317/KHLDGNet**

***Index Terms*—Domain generalization (DG), hyperspectral images (HSIs), knowledge-driven learning, vision–language alignment, wetland land-covered classification.**

I. Introduction

WETLANDS rank among the most productive yet vulnerable ecosystems on Earth, playing critical roles in maintaining biodiversity, regulating climate, and purifying water [1], [2], [3], [4]. Accurate and timely monitoring of wetland resources is therefore a prerequisite for their scientific conservation and sustainable management. In recent decades, remote sensing (RS) technology has become an indispensable tool for large-scale wetland mapping. In particular, hyperspectral images (HSIs), characterized by their exceptionally high spectral resolution, enable the discrimination of subtle spectral signatures of ground objects, offering unique advantages for fine-scale wetland classification [5], [6], [7].

Currently, deep learning-based methods, especially convolutional neural networks (CNNs), have achieved significant success in wetland classification of HSI [8], [9]. Xie et al. [10] proposed a multilayer global spectral spatial attention network (MGSSAN) that extracts discriminative features through a dual-branch CNN combined with a global attention mechanism, and optimizes the initial classification probability using extended random walks. Li et al. [11] proposed a feature-guided dynamic graph convolutional network (FG-DGCN), which adaptively partitions regions through a learnable superpixel generation module and fuses spectral and spatial correlations using a feature-guided dynamic adjacency matrix. Dong et al. [12] proposed a hardness-aware prototypical contrastive learning (HAPC) method, which effectively solves the balance problem between fixed temperature parameters and difficult negative sample identification in self-supervised contrastive learning through spatial spectral ensemble data augmentation and a hardness perception temperature weighting strategy. Meanwhile, to alleviate the heavy reliance on large-scale labeled data, few-shot learning has emerged as a promising direction for HSI classification, aiming to recognize novel categories with only a handful of labeled samples per class [13], [14]. However, the vast majority of existing methods are trained and tested within a single scene, relying on an ideal assumption: the training set and the test set originate from the same scene and are independently and identically distributed. In reality, field surveys and sampling in wetlands are challenging, and obtaining sample labels is often difficult. This severely hinders the applicability of existing methods to real-world wetland classification problems, as many wetland scenes even lack annotations entirely. Furthermore, the acquisition of hyperspectral imagery is inevitably influenced by factors such as sensor discrepancies, imaging conditions, and geographical

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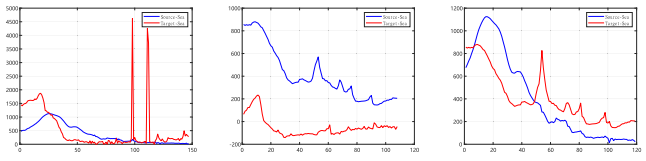
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(a) (b) (c)

Fig. 1. Spectral drift between different scenes. (a) Same region with different sensors. (b) Same sensor with different acquisition times. (c) Same sensor across different geographical scenes.

variations. These factors lead to significant spectral shifts and distribution differences for the same wetland category across different scenes or times, a phenomenon known as the domain shift problem [15]. As illustrated in Fig. 1, the spectral curves of the same category exhibit marked inconsistencies between the source and target domains. Consequently, directly applying a model trained on the source domain to the target domain typically results in a drastic decline in classification performance and a severe lack of generalization capability.

To address the domain shift challenge, methods based on transfer learning have been extensively studied [16]. Among these, domain adaptation (DA) methods aim to learn domain-invariant feature representations by leveraging labeled source domain data together with unlabeled target domain data during training, thereby reducing the discrepancy between domains [17]. Ye et al. [18] proposed a dictionary learning-based feature-level DA method, which effectively improves the accuracy of cross-scene classification by mapping the spectral features of the source and target scenes to a shared low-dimensional embedding space to align distributions, and combining sparse logistic regression for classification. Tang et al. [19] proposed an unsupervised joint adversarial DA (UJADA) architecture, which utilizes domain adversarial and dual classifier adversarial modules to achieve domain-level and class-level feature alignment in a unified adversarial learning process, in order to improve the cross-scene classification performance of HSIs. Zhang et al. [20] proposed a topology and semantic information transfer network (TSTnet) to address the issue of existing methods ignoring data topology. TSTnet assists in distribution alignment by aligning the topology relationship between the source and target domains through graph optimal transmission, and integrates the advantages of graph convolutional networks and CNNs to improve cross-scene classification performance. Huang et al. [21] proposed a spatial–spectral weighted adversarial DA network (SSWADA), which achieves precise alignment of feature distributions between source and target domains and improves classification performance by combining 2-D–3-D convolutional feature extractors, source instance weighted discriminators, and a multiclassifier structure. Cui et al. [22] proposed a mask-guided and confidence-driven unsupervised DA (MCUDA) method, which dynamically optimizes input data through point-level learnable masks to suppress interference, and combines a pseudolabel generation strategy based on confidence learning to utilize instance-level differences in the target domain, thus achieving better performance than existing methods in cross-scene hyperspectral classification. Wang et al. [23] proposed a cross-domain feature extraction and category decision collaborative optimization network (FECD), which effectively enhances the model's cross-scene discrimination ability and optimizes the classification boundary through dynamic mask adaptive feature discovery, Mamba and ViT dual-stream feature extraction, and dual-stream classifier collaborative optimization. However, the fundamental limitation of DA methods lies in their requirement to access target domain data. In practical wetland monitoring applications, images of target domains often lack labels and may even be entirely unavailable in advance, which severely restricts the practical applicability of DA methods.

In contrast, domain generalization (DG) represents a paradigm that is more aligned with practical scenes yet more challenging. DG methods are trained using only one or several labeled source domains, aiming to learn a model that can generalize directly to any unseen target domain [24], [25]. Zhao et al. [26] proposed a method based on symmetric encoding–decoding generative networks, which constructs local joint features through style randomization and balances domain-specific and domain-invariant features by combining supervised contrastive learning and adversarial contrastive regularization. Zhang et al. [27] proposed a single-source domain expansion network (SDEnet) that generates an extension domain with diverse spectral spatial features, utilizes morphological knowledge as domain-invariant information, and combines supervised contrastive learning and adversarial training to learn category-level domain-invariant representations. This achieves effective generalization to unknown target domains when trained only on source domain data. Wang et al. [28] proposed a two-stage domain alignment single-source DG network (TSDAnet). By introducing spectral learning branches to optimize generation, a dual projection pyramid feature fusion network was designed to enhance discrimination, and a two-stage alignment loss was constructed to more comprehensively eliminate domain offset, thereby improving the model's generalization classification ability for unseen target domains. Peng et al. [29] proposed a disentanglement-inspired single-source domain-generalization network (DSDGnet), which extends the source domain through a style transfer module and designs a progressive decoupling module and a domain combination module to separate domain-invariant features from domain-specific features, effectively alleviating spectral heterogeneity. Gao et al. [30] proposed an invariant semantic DG shuffle network (ISDGS) based on generative adversarial networks and contrastive learning strategies, aiming to achieve effective generalization of unseen scenes without accessing the target domain by maintaining domain-invariant semantics and optimizing the training process. Zhao et al. [31] proposed a DG network that combines adversarial decoupling and unbiased semantic extension. By decoupling causal components related to labels and synthesizing extended domains that preserve core semantics, the network aims to eliminate spurious correlations in data augmentation. Existing DG approaches primarily operate at the visual level, either by simulating domain variations through data augmentation or by learning more generalizable features through techniques such as meta-learning and style transfer

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[32]. However, most of these methods overlook another crucial dimension: semantic information imbued with rich prior knowledge. Geographic entities are not merely defined by their visual appearance but also encompass profound semantic connotations, such as their spatial distribution patterns, ecological functions, and the mechanisms governing their formation and evolution. This semantic knowledge is often shared and invariant across different domains. If effectively integrated into the model learning process, it holds the potential to serve as a powerful anchor for guiding the model toward achieving robust generalization.

Recently, the success of large-scale vision–language foundation models (e.g., CLIP [33]) has revealed the immense potential of leveraging natural language supervision to learn powerful visual representations. This type of model embeds features from different modalities into a shared semantic space through contrastive learning of massive image–text pairs. It not only learns high-quality visual representations, but its alignment mechanism also provides a natural and interpretable decision path: the model's predictions can be obtained by calculating the semantic similarity between visual features and a series of textual descriptions, such as category names or attributes, making the "why the model made this judgment" traceable and understandable. Inspired by this, researchers in the field of RS have begun to explore the use of visual–language alignment paradigms for tasks such as land cover classification. Zhang et al. [34] proposed a language-aware DG network (LDGnet), which constructs a cross-domain shared semantic space by extracting multigranularity text features and aligns visual and language modalities using supervised contrastive learning. Wang et al. [35] proposed an explicit high-level semantic network (EHSnet) for DG. By designing a dual residual encoder to extract spatial spectral features of HSIs and aligning them with multilevel text semantics in a shared space, the model is endowed with zero-sample generalization ability to address cross-scene classification challenges. Wang et al. [36] proposed a text-driven adaptive semantic alignment network (TASA-Net) that combines handwritten template prompts with visually guided fine-grained semantic prompts, and utilizes a dual door adaptive mechanism and soft alignment strategy to achieve cross-modal alignment in a shared semantic space. However, existing vision–language methods, despite their effectiveness, rely on relatively simplistic textual descriptions, typically category names or isolated attributes. They lack a structured hierarchical formulation that systematically characterizes geographic categories from conceptual essence and defining rules to diverse instantiations and, most importantly, underlying causal mechanisms. Consequently, these methods face three limitations: 1) flat semantics without multilevel reasoning; 2) no causal understanding to anchor domain-invariant features; and 3) black-box alignment without interpretable decision tracing.

In response to the above challenges and the pressing call for advancing explainable geographic artificial intelligence (XGeoAI), this article proposes a novel knowledge-driven hierarchical language-aware DG network (KHLDGNet) for interpretable cross-scene wetland HSI classification. The core idea is to inject structured, multilevel geographic prior knowledge in textual form into the DG framework, constructing a cross-domain shared, semantics-rich space to guide visual feature learning toward more robust and explainable generalization. The main contributions of this article are summarized as follows.

1) A structured four-level geographic knowledge hierarchy is constructed. Encompassing concept, rule, instance, and causality, this hierarchy transforms a geographical cognitive framework into computable semantic constraints, providing the model with comprehensive semantic guidance from fundamental attributes to dynamic mechanisms.
2) A style-enhanced visual encoder is designed. A MixStyle module is integrated into the 3D-CNN backbone to implicitly increase feature-level style diversity, thereby improving the model's adaptability to unseen domain-specific spectral and textural variations.
3) A traceable decision path is achieved through hierarchical contrastive learning. By projecting HSI visual features into a semantic space constructed from hierarchical text, the model enables classification decisions to be traced back to specific textual descriptions, providing intuitive and semantically grounded explanations.
4) Comprehensive experiments are conducted on three real-world cross-scene coastal wetland HSI datasets. The results demonstrate that KHLDGNet significantly outperforms existing DG methods. Visualization and case studies concretely illustrate how the model leverages different levels of textual knowledge for decision-making, affirming its value in advancing XGeoAI.

The remainder of this article is organized as follows. Section II elaborates on the proposed methodology. Section III describes the datasets, experimental setup, and presents a detailed analysis of the results. Finally, Section IV concludes this article and discusses potential future work.

II. Methodology

A. Overall Architecture

The overall architecture of the proposed KHLDGnet is illustrated in Fig. 2. It is a multimodal learning framework designed to achieve interpretable cross-scene hyperspectral wetland classification by integrating structured textual knowledge with visual feature learning. The architecture consists of four key components: 1) a hierarchical text knowledge construction module, which generates four levels of semantic descriptions for each land cover category; 2) a style-enhanced visual encoder, which extracts robust spatial–spectral features while implicitly augmenting style diversity via a MixStyle module; 3) a Transformer-based text encoder, which encodes hierarchical textual descriptions into dense semantic embeddings; and 4) a visual–language alignment module, which projects both visual and textual features into a shared semantic space and aligns them through supervised contrastive learning.

During training, HSI patches are fed into the visual encoder to extract deep features, while the corresponding hierarchical text descriptions are processed by the text encoder. The

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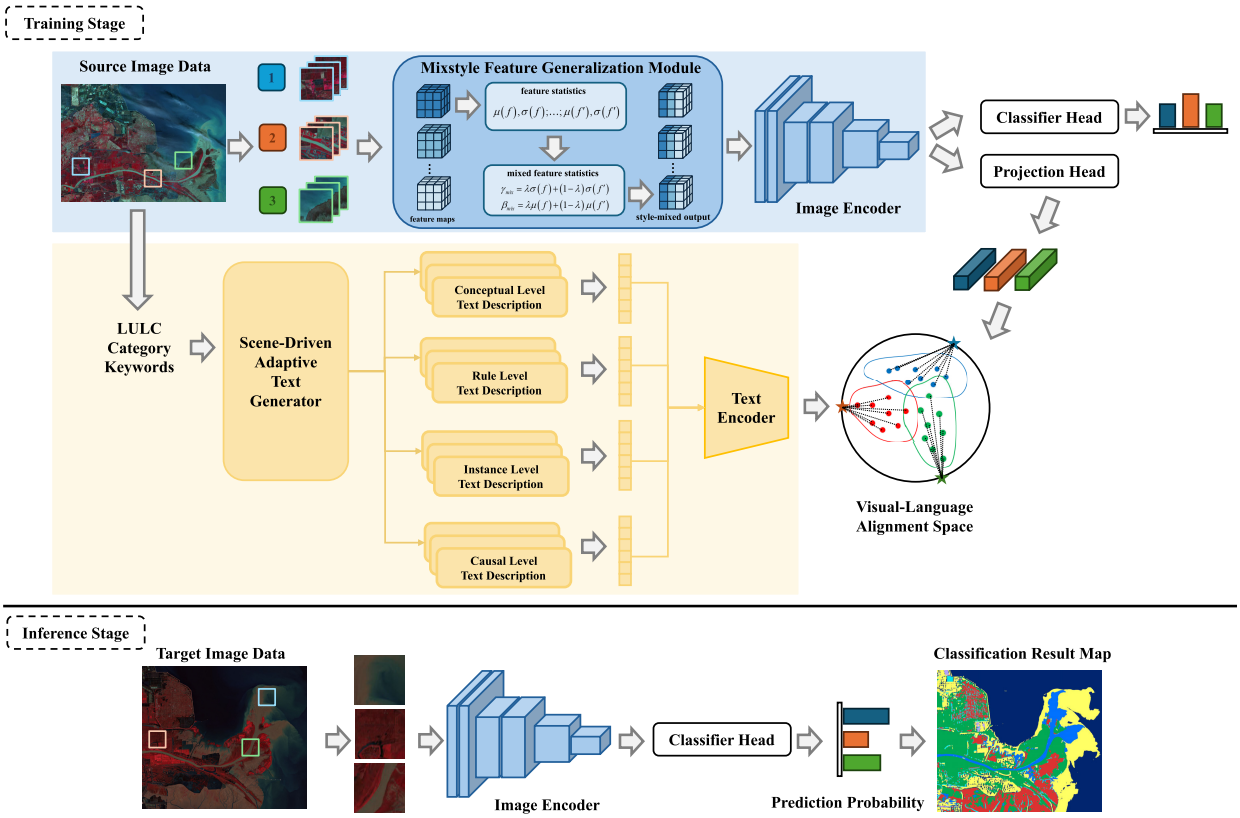


Fig. 2. Overall flowchart of the proposed KHLDGnet.

visual–language alignment module then enforces semantic consistency across the four textual levels, guiding the visual features to align with domain-invariant semantic concepts. The overall loss function combines a conventional cross-entropy classification loss on the source domain with a weighted hierarchical alignment loss, jointly optimized in an end-to-end manner.

Distinct from prior vision–language DG works (e.g., LDGnet [34]) that employ flat category-name prompts and lack interpretability, our KHLDGNet introduces three key innovations: 1) a four-level geographic knowledge hierarchy that provides structured semantic guidance; 2) a MixStyle-enhanced visual encoder that explicitly addresses cross-scene spectral-textural variations; and 3) a traceable decision mechanism via hierarchical contrastive learning. During inference, only the trained visual encoder and a lightweight classification head are retained, ensuring efficient prediction without any additional computational overhead from the text modality. This design not only enhances the model's generalization ability to unseen wetland scenes but also provides an intrinsically interpretable decision pathway, where classification outcomes can be traced back to specific levels of semantic descriptions, thereby advancing the paradigm of explainable GeoAI.

B. Hierarchical Text Knowledge Construction

To overcome the limitation of flat semantic prompts in existing vision–language methods, which lack structured reasoning and causal understanding, we design a four-level knowledge hierarchy that provides comprehensive semantic guidance from fundamental attributes to causal mechanisms. Prior vision–language works in RS (e.g., LDGnet and EHSnet) typically employ simple textual prompts such as category names ("an HSI of wetland") or isolated attributes ("dark color, irregular shape"). While effective to some extent, these flat descriptions lack structured reasoning about why a category exhibits certain visual characteristics. To systematically infuse structured semantic priors into the model and guide it toward learning domain-invariant representations, we design a hierarchical textual knowledge construction framework. Moving beyond simple category names or attributes used in prior vision–language works, this framework decomposes the semantic knowledge of each LULC category into four progressively deeper and more abstract levels: conceptual, rule, instance, and causal [37], [38]. This design forms a coherent semantic modeling hierarchy, aiming to provide the model with a complete understanding from basic identification to functional and generative mechanisms.

1) Conceptual Level Text Description: This level provides the fundamental semantic identifier, answering "what" the category is. It is defined as a template: "an HSI of [LULC category]." This establishes the basic name-to-visual association and serves as the anchor for supervised learning, analogous to the class label alignment in foundational models like CLIP.

2) Rule Level Text Description: This level abstracts the core, discriminative attributes that are stable across domains, focusing on "how" the category is defined beyond its spe-

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TABLE I
COMPARISON OF TEXTUAL KNOWLEDGE STRUCTURES IN VISION–LANGUAGE DG METHODS

Feature	LDGNet [34]	EHSnet [35]	KHLDGNet(Ours)
textual knowledge structure	category names/attributes	Handcrafted templates + visual prompts	Hierarchical four-level
expert knowledge	general knowledge	expert-crafted handcrafted templates	geographic expert knowledge hierarchy + generative AI
visual prompt tuning		Yes	No
interpretable path	No	No	Yes

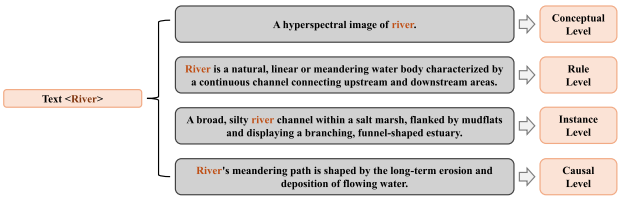


Fig. 3. Examples of text for one LULC category.

cific appearance. It describes inherent and often invariant characteristics—such as geometric shapes, structural patterns, or spatial relationships—derived from domain expertise. This guides the model to learn essential features resilient to superficial stylistic changes.

3) Instance-Level Text Description: This level offers contextualized and visually rich exemplars, illustrating "what it looks like" in a typical scene. It describes not only the object itself but also its common state, surrounding environment, and associated visual details such as color, texture, and layout. Serving as a form of semantic data augmentation, it expands the model's understanding of intraclass diversity and improves robustness in complex scenes.

4) Causal-Level Text Description: This level explains the underlying mechanisms or functional principles, answering "why" the category exhibits certain rule-based or instance-based characteristics. It connects visual attributes to underlying factors such as physical processes, human intentions, or ecological functions. For example, it encapsulates the principle that the grid layout of buildings results from urban planning. This provides the deepest semantic constraints, promoting the learning of highly transferable and explanatory features.

As shown in Fig. 3, each land cover category is assigned four corresponding texts. This hierarchical knowledge injection enables the model to go beyond shallow visual pattern matching and learn more generalized and interpretable geographic concept essence. Together, these four levels form a closed-loop semantic guidance chain: the conceptual level anchors the category, the rule level defines its essential attributes, the instance level illustrates its diverse appearances, and the causal level reveals its underlying principles. During training, multilevel alignment forces the visual encoder to reconcile its features with this structured semantic hierarchy, thereby compressing its dependency on domain-specific superficial cues and promoting generalization through domain-invariant semantic grounding.

To better understand the differences between our proposed KHLDGNet and existing vision–language DG methods, Table I compares LDGNet [34], EHSnet [35], and our method across four key dimensions: textual knowledge structure, expert knowledge involvement, visual prompt tuning, and interpretable decision path. As shown, prior methods rely on either simple category names/attributes or handcrafted templates combined with visual prompts. In contrast, KHLDGNet introduces a hierarchical four-level knowledge system grounded in geographic expert knowledge and augmented by generative AI. Moreover, our framework provides an intrinsically interpretable decision path by tracing predictions back to specific semantic descriptions, a feature absent in existing approaches. The effectiveness of this hierarchical design is validated in our ablation study, where removing any semantic level leads to clear performance degradation.

C. Visual Encoder

1) MixStyle Feature Generalization Module: To address the spectral-textural style variations in HSI caused by different sensors and seasons, we introduce the MixStyle module to implicitly enhance feature-level style diversity. In HSI cross-scene wetland classification, significant style differences often occur between the source domain and the target domain due to differences in imaging sensors, shooting seasons, lighting conditions, and geographical environments. This difference is directly manifested as a shift in feature distribution and is one of the key factors leading to a decrease in model generalization ability. Although hierarchical language description can constrain the cross-domain consistency of features at the high-level semantic level, effectively modeling and adapting to this style change in low-level visual features is equally crucial for improving overall generalization performance. To this end, we introduced the MixStyle feature generalization (MixStyle) module at the front end of the visual encoder, aiming to implicitly increase the style diversity of the training data at the feature level, so that the model can learn visual representations that are more robust to interdomain style changes without introducing any additional computational overhead in the inference stage. The structural schematic of this module is shown in Fig. 4. Specifically, given the input feature map $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$, first calculate its channel level mean μ and standard deviation σ

$$\mu_c = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} F_{c,h,w} \tag{1}$$

$$\sigma_c = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(F_{c,h,w}-\mu_c\right)^2+\epsilon} \tag{2}$$

where ϵ is the numerical stability constant. Next, we perform instance normalization on the features

$$\hat{F} = \frac{F-\mu}{\sigma}. \tag{3}$$

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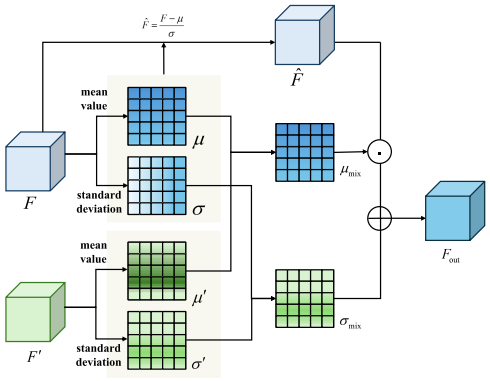


Fig. 4. Structure of MixStyle feature generalization module.

During the training phase, we use probability p to determine whether to perform style mixing on the current batch. If activated, randomly select another set of samples from the current batch to obtain their statistical values μ' and σ', and sample the mixing coefficient $\lambda \sim \text{Beta}(\alpha, \alpha)$

$$\mu_{\text{mix}} = \lambda\mu + (1-\lambda)\mu', \quad \sigma_{\text{mix}} = \lambda\sigma + (1-\lambda)\sigma'. \tag{4}$$

Finally, the feature output after style mixing is

$$F_{\text{out}} = \widehat{F} \odot \sigma_{\text{mix}} + \mu_{\text{mix}}. \tag{5}$$

By using a mixed-style feature generation module, it can be seen as generating a series of "intermediate styles" between the source domain and the virtual domain in the feature space, thereby enhancing the model's adaptability to spectral and texture changes. In the cross-scene wetland classification task, we specifically inserted a mixed-style feature generation module before the deep feature extraction network. This is because shallow features contain more spectral and texture style information, while deep features are more inclined toward semantic content. By disturbing the style statistics at this level, we can simulate the spectral response changes under different sensors, seasons, and lighting conditions without changing the wetland category semantics, so as to improve the generalization performance of the model in unknown wetland scenes. By integrating a mixed-style feature generation module into our hierarchical language perception DG framework, we not only enhance the robustness of visual encoders to style changes, but also provide more discriminative and generalizable feature representations for subsequent visual–language alignment.

2) Image Encoder: HSIs have a 3-D data structure, containing hundreds of spectral bands and rich spatial and spectral information. For this purpose, we adopt a deep 3-D residual CNN (3D-RCNN) as the backbone network for visual feature extraction, as shown in Fig. 5. This network extracts discriminative spatial spectral features layer by layer through multiple residual blocks and pooling layers, providing robust visual representations for subsequent visual–language alignment. The forward propagation process can be expressed as

$$\begin{aligned} x_1 &= \text{ReLU}\left(\text{BN}\left(\text{Conv3D}\left(x; \mathbf{W}_1, \mathbf{b}_1\right)\right)\right) \\ x_2 &= \text{ReLU}\left(\text{BN}\left(\text{Conv3D}\left(x_1; \mathbf{W}_2, \mathbf{b}_2\right)\right)\right) \\ x_3 &= \text{Conv3D}\left(x_2; \mathbf{W}_3, \mathbf{b}_3\right) \end{aligned}$$

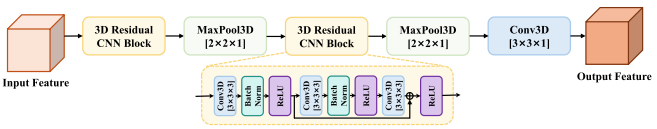


Fig. 5. Structure of deep feature extraction network.

$$F_{out} = \text{ReLU}\left(x_1 + x_3\right). \tag{6}$$

This network structure fully captures the local spectral changes and global spatial context in HSIs, effectively alleviates the problem of gradient vanishing, and enhances the expression ability of features, laying a solid visual foundation for subsequent multimodal alignment and domain-invariant representation learning. We flatten the final feature map and feed it separately into the classification head and projection head

$$\begin{aligned} \mathbf{z}_{\text{cls}} &= \mathbf{W}_{\text{cls}} \cdot \text{Flatten}(\mathbf{F}_{\text{out}}) + \mathbf{b}_{\text{cls}} \\ \mathbf{z}_{\text{proj}} &= \mathbf{W}_{\text{proj}} \cdot \text{Flatten}\left(\mathbf{F}_{\text{out}}\right) \end{aligned} \tag{7}$$

where $\mathbf{z}_{\text{cls}}$ is used to calculate the cross-entropy classification loss, and $\mathbf{z}_{\text{proj}}$ is used for subsequent visual–language alignment. The definition of source domain classification loss is

$$L_{\text{SD}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log\left(\text{Softmax}\left(\mathbf{z}_{\text{cls}}^{(i)}\right)_c\right). \tag{8}$$

D. Text Encoder

In the text encoding stage, we adopt a text encoder architecture consistent with the Transformer [39] language model modified by Radford et al. [40]. Ensure the reliability and comparability of text feature extraction. This encoder is built on a pretrained Transformer architecture and can transform hierarchical text knowledge into dense semantic vector representations, providing a high-quality text feature foundation for subsequent visual–language alignment. The text encoder starts with a basic size of 33-M parameters, consists of three layers, a 512-wide model, and has eight attention heads. Similar to CLIP, the text is represented by a Transformer using lowercase byte pair encoding (BPE) with a vocabulary size of 49 152 [41]. For computational efficiency, the maximum sequence length is limited to 76. The text encoder is initialized with CLIP's pretrained weights and fine-tuned during training. To mitigate overfitting under limited samples, we apply early stopping and a weight decay of $1e^{-4}$. The core of a text encoder is the Transformer encoder, which is composed of multiple layers of self-attention blocks stacked together. Each attention block includes a multihead self-attention mechanism, layer normalization, and a feedforward neural network. The specific implementation is as follows:

$$\begin{aligned} \text{MultiHeadAttention}(\mathbf{H}) &= \text{Concat}\left(\text{head}_1, \ldots, \text{head}_h\right)\mathbf{W}_O \\ \text{head}_j &= \text{Attention}\left(\mathbf{H}\mathbf{W}_Q^j, \mathbf{H}\mathbf{W}_K^j, \mathbf{H}\mathbf{W}_V^j\right) \\ \text{Attention}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) &= \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V} \end{aligned} \tag{9}$$

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where $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$, and $\mathbf{W}_O$ are a learnable projection, and $\mathbf{H}$ is the number of attention heads d_k is the dimension of the key vector.

For each LULC category k with four levels of text description $T_K = \{T_k^{\text{conc}}, T_k^{\text{rule}}, T_k^{\text{inst}}, T_k^{\text{caus}}\}$, the text encoding process first converts the text sequence into word vectors and adds learnable positional encoding

$$E_k{}^{\ell} = \text{Embedding}\left(T_k{}^{\ell}\right) + \text{PositionalEmbedding} \tag{10}$$

where Embedding() is the word embedding layer, and PositionalEmbedding is the learnable position encoding matrix. After that, semantic features are extracted through a multilevel Transformer encoder after adjusting the tensor dimension

$$H_k^{\ell,(i)} = \text{Transformer}^{(i)}\left(E_k^{\ell,(i-1)}\right), \quad i = 1, \ldots, N. \tag{11}$$

Next, perform layer normalization on the Transformer output

$$x = \text{LayerNorm}\left(H_k^{\ell}\right). \tag{12}$$

Finally, extract the feature vectors corresponding to the end-of-sequence (EOS) markers in each text sequence and map them to the target dimension through a projection layer. The EOS tag is obtained through text.argmax (dim = −1), representing the maximum index position in each sequence

$$\begin{aligned} \text{eos_indices} &= \arg\max(\text{text}, \text{dim} = -1) \\ x_{\text{eos}} &= x\left[\text{torch.arange}\left(x.\text{shape}[0]\right), \text{eos_indices}\right] \\ t &= x_{\text{eos}} \cdot \mathbf{W}_{\text{proj}} \end{aligned} \tag{13}$$

where $\mathbf{W}_{\text{proj}}$ is the text projection matrix, which aims to project the feature dimension from the Transformer width to the dimension d_t aligned with the visual features.

During the training phase, the text encoder and visual encoder jointly participate in forward propagation, aligning text features with visual features through contrastive learning loss. In reasoning, the text encoder is completely removed, and the model relies only on the features extracted by the visual encoder for classification decisions, achieving zero additional computational overhead. This design enables the model to fully utilize the rich semantic knowledge of large-scale pretrained language models while maintaining efficient inference in practical RS interpretation tasks, providing an effective technical path for interpretable GeoAI.

E. Visual–Language Alignment

The visual–language alignment module is the core of this method for achieving cross-DG and interpretable decision-making. This module aims to project the visual features output by the deep feature extraction network in the visual encoder and the structured hierarchical language features generated by the text encoder into a cross-domain shared semantic space, and achieve fine-grained feature alignment through supervised contrastive learning. The key is to regard language knowledge as a universal semantic bridge connecting different domains, thereby guiding the visual model to learn feature representations that are robust to domain changes and have semantic interpretability.

Given a mini-batch of visual features $\mathbf{V} = \{\mathbf{v}_i\}_{i=1}^{B}$ and their corresponding category labels $\mathbf{y} = \{\mathbf{y}_i\}_{i=1}^{B}$, as well as a set of text features $\mathbf{T}_c = \{t_c^{\text{conc}}, t_c^{\text{rule}}, t_c^{\text{inst}}, t_c^{\text{caus}}\}$ at four levels (concept, rule, instance, and causality) for each category c. The goal of alignment is to narrow the distance between visual features belonging to the same category and all hierarchical text features of that category in the semantic space, while pushing the distance between features of different categories. We use supervised contrastive loss to achieve the above objectives. For each level, the definition of image-to-text contrast loss is as follows:where $P_{txt}^{(l)}(i)$ is the set of positive textual samples for visual feature $\mathbf{v}_i$ belonging to class y_i, containing the level-ℓ textual features corresponding to class y_i; $A_{txt}^{(l)}(i)$ is the set of negative textual samples, containing all textual features (at the same level) in the batch that do not belong to class y_i; and τ is a temperature hyperparameter that controls the sharpness of the similarity distribution.

Similarly, the text-to-image contrastive loss $L_{\text{txt}\to\text{img}}^{(\ell)}$ is defined symmetrically, centering on each textual feature to pull it closer to visual features of the same class and push it away from visual features of different classes. The total alignment loss for each level is the sum of the image-to-text and text-to-image losses at that level

$$L_{\text{align}}^{(\ell)} = L_{\text{img}\to\text{txt}}^{(\ell)} + L_{\text{txt}\to\text{img}}^{(\ell)} \tag{15}$$

Finally, the overall loss for the hierarchical visual-semantic alignment module is the weighted sum of the losses across all four levels

$$L_{\text{VL-Align}} = \sum_{\ell \in \text{conc, rule, inst, caus}} \beta_\ell \cdot L_{\text{align}}^{(\ell)} \tag{16}$$

where β_ℓ is a hyperparameter used to balance the contribution of semantic supervision from different levels. Specifically, we set $\beta_{conc} = 0.4$, $\beta_{rule} = 0.3$, $\beta_{inst} = 0.2$, and $\beta_{caus} = 0.1$ based on empirical validation on a small held-out set from the source domain. This configuration is designed to emphasize the fundamental role of concept-based descriptions in establishing a stable and domain-invariant semantic foundation, while still incorporating fine-grained semantic guidance from rule-based, instance-based, and causal-based descriptions in a balanced manner. By optimizing this loss function, the model is forced to cluster multimodal features of the same class in adjacent regions of the semantic space.

The introduction of hierarchical text, especially rule-based and causal descriptions, provides deep, domain-invariant semantic constraints that go far beyond superficial appearances. This not only enhances the model's generalization

$$L_{\text{img}\to\text{txt}}^{(\ell)} = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{\left|P_{\text{txt}}^{(\ell)}(i)\right|}\sum_{p\in P_{\text{txt}}^{(\ell)}(i)}\log\frac{\exp\left(\mathbf{v}i\cdot\mathbf{t}_p^{(\ell)}/\tau\right)}{\sum a\in A_{\text{txt}}^{(\ell)}(i)\exp\left(\mathbf{v}_i\cdot\mathbf{t}_a^{(\ell)}/\tau\right)} \tag{14}$$

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ability on the target domain but also allows the model's visual classification decisions to be traced and explained through their proximity to different levels of semantic descriptions, providing a key mechanism for achieving intrinsically XGeoAI. During inference, only the trained visual encoder and classification head are used to classify target domain images, while the text encoder and alignment module are not involved in computation, thus introducing no additional overhead.

F. Network Training and Inference

The goal of network training is to simultaneously optimize the parameters of the visual encoder, text encoder, and visual semantic alignment module to learn models with strong generalization ability and inherent interpretability. The overall loss function L_{Total} of the model consists of source domain classification loss and hierarchical visual–language alignment loss

$$L_{\text{Total}} = L_{\text{SD}} + \lambda \cdot L_{\text{VL-Align}} \tag{17}$$

The specific implementation steps of the proposed algorithm are summarized as Algorithm 1.

Algorithm 1 KHLDGNet Training Procedure

Input: Source domain hyperspectral patches $\mathbf{X}_s$, labels $\mathbf{Y}_s$, Hierarchical text descriptions $\{\mathbf{T}_k\}$ for each class k, Hyperparameters: λ, β_{conc}, β_{rule}, β_{inst}, β_{caus}, p, α, τ;
Output: Trained visual encoder θ_v and classification head θ_{cls}.

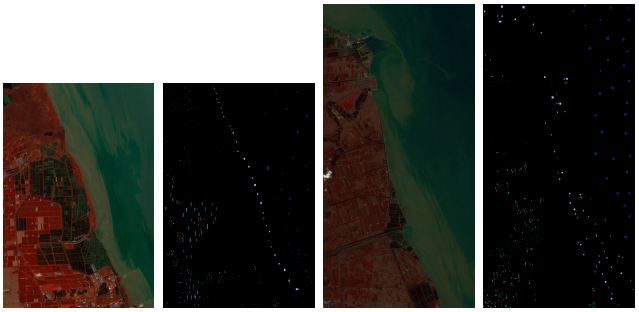
Initialize visual encoder θ_v, text encoder θ_t, classification head θ_{cls}
while not converged **do**:
  Sample a mini-batch $(\mathbf{X}_b, \mathbf{Y}_b)$ from source domain
  Step 1: Visual Feature Extraction
  $\mathbf{X} \leftarrow \text{Mixstyle}(\mathbf{X}_b)$ \\ Eq. (1)–(5)
  $\mathbf{F}_v \leftarrow \text{VisualEncoder}(\mathbf{X}; \theta_v)$ \\ Eq. (6)–(6)
  $\mathbf{Z}_{\text{cls}} \leftarrow \text{ClassificationHead}(\mathbf{F}_v; \theta_{\text{cls}})$
  $L_{\text{SD}} \leftarrow \text{CrossEntropy}(\mathbf{Z}_{\text{cls}}, \mathbf{Y}_b)$ \\ Eq. (6)
  Step 2: Text Feature Extraction
  $\mathbf{T}_b \leftarrow \{\mathbf{T}_y \mid y \in \mathbf{Y}_b\}$ \\4-level texts per sample
  $\mathbf{F}_t \leftarrow \text{TextEncoder}(\mathbf{T}_b; \theta_t)$ \\ Eq. (10)–(13)
  Step 3: Hierarchical Visual-Language Alignment
  $L_{\text{VL-Align}} \leftarrow 0$
  for $\ell \in \{\text{conc, rule, inst, caus}\}$ **do**
  $L_{\text{align}}^{(\ell)} \leftarrow L_{\text{img}\to\text{txt}}^{(\ell)} + L_{\text{txt}\to\text{img}}^{(\ell)}$ \\ Eq. (14)–(15)
  $L_{\text{VL-Align}} \leftarrow L_{\text{VL-Align}} + \beta_\ell \cdot L_{\text{align}}^{(\ell)}$
  end for
  Step 4: Total Loss & Backward
  $L_{\text{Total}} \leftarrow L_{\text{SD}} + \lambda \cdot L_{\text{VL-Align}}$ \\ Eq. (17)
  Update θ_v, θ_t, θ_{cls} by minimizing L_{Total}
end while
Return θ_v, θ_{cls}

In the model inference stage, only the trained visual encoder and classification head need to be used. For any input target domain image block, after extracting features through a visual encoder, the classification head directly outputs category predictions. The text encoder and alignment module, which play a crucial role in training, are completely removed during inference without introducing any additional computational or storage overhead, achieving zero-cost DG and semantic knowledge injection. The specific implementation steps of the inference stage are summarized as Algorithm 2.

Algorithm 2 KHLDGNet Inference Procedure

Input: Target domain hyperspectral patch $\mathbf{X}_t$, Trained visual encoder θ_v and classification head θ_{cls};
Output: Predicted class label $\hat{y}$.

Extract visual features: $\mathbf{F}_t \leftarrow \text{VisualEncoder}(\mathbf{X}_t; \theta_v)$
Predict class: $\mathbf{Z}_t \leftarrow \text{ClassificationHead}(\mathbf{F}_t; \theta_{\text{cls}})$
$\hat{y} \leftarrow \arg\max(\mathbf{Z}_t)$
Return $\hat{y}$



(a) (b) (c) (d)

Fig. 6. YC scene. (a) Pseudocolor image of ZY1-02D Yancheng. (b) GT map of ZY1-02D Yancheng. (c) Pseudocolor image of GF-5 Yancheng. (d) GT map of GF-5 Yancheng.

III. Experiments and Analysis

A. Data Description

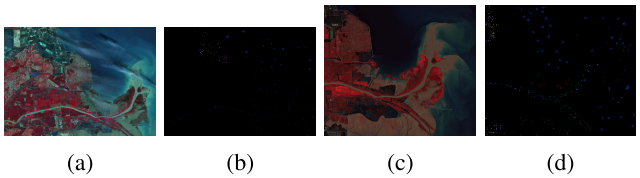
To systematically validate the effectiveness of the proposed KHLDGNet for cross-scene hyperspectral wetland classification, we conducted experiments on three real-world wetland HSI datasets with typical domain shift characteristics. These datasets represent three common types of distribution shifts encountered in real-world applications: sensor discrepancy, temporal variation, and geographical divergence. Together, they form a comprehensive cross-domain testbed for evaluating the DG capability of the model.

1) *Sensor Discrepancy Scenario (Yancheng (YC) Dataset):* This dataset is designed to evaluate the model's robustness to domain shifts caused by sensor discrepancy. It consists of two images covering the same geographical area, acquired by the advanced HSIr (AHSI) onboard the ZY1-02D and GF-5 satellites, respectively. They are used as the source and target domains to construct a cross-domain classification task of the same region, using different sensors. The detailed categories and sample size distribution are shown in Table II. The pseudocolor image and ground truth (GT) map are shown in Fig. 6.
2) *Temporal Variation Scenario Huanghekou (HHK) Dataset):* This dataset is used to examine the model's adaptability to domain shifts induced by temporal variation. The data are both from the ZY1-02D satellite sensor but were captured at different times (June 2020 and September 2021). This reflects the data distribution

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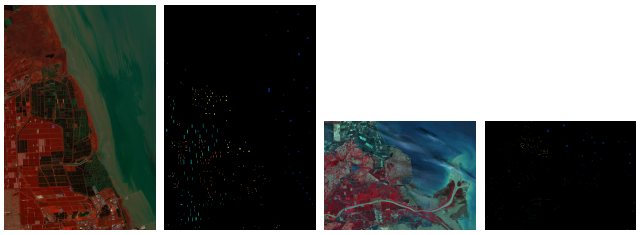
TABLE II
CATEGORIES AND CORRESPONDING QUANTITIES OF SAMPLES

YC (sensor discrepancy)				HHK (temporal variation)				YC-HHK (geographical divergence)			
No.	Class	Source	Target	No.	Class	Source	Target	No.	Class	Source	Target
1	Architecture	451	360	1	Reed	310	1057	1	Architecture	451	398
2	River	214	217	2	Salt Flat Filtration Pond	247	512	2	Paddy	3026	508
3	Reed	244	132	3	Salt Flat Evaporation Pond	300	260	3	Fallow land	650	459
4	Paddy	3026	832	4	Salt Flat	306	412	4	Fish pond	920	124
5	Fallow land	650	234	5	Suaeda	218	418	5	Sea	2076	4694
6	Sea	2076	2395	6	River	584	941	6	Salt pond	901	853
7	Offshore water	1558	1305	7	Sea	4694	10239				
				8	Tide Ditch	67	248				
Total		8219	5475	Total		6726	14087	Total		8024	7036



(a) (b) (c) (d)

Fig. 7. HHK scene. (a) Pseudocolor image of ZY1-02D Huanghekou (2020). (b) GT map of ZY1-02D Huanghekou (2020). (c) Pseudocolor image of ZY1-02D Huanghekou (2021). (d) GT map of ZY1-02D Huanghekou (2021).



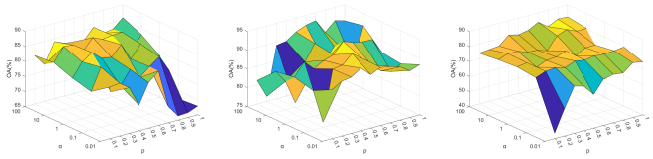
(a) (b) (c) (d)

Fig. 8. YC-HHK scene. (a) Pseudocolor image of ZY1-02D Yancheng. (b) GT map of ZY1-02D Yancheng. (c) Pseudocolor image of ZY1-02D Huanghekou (2020). (d) GT map of ZY1-02D Huanghekou (2020).

drift due to seasonal changes, illumination variations, and vegetation phenology under the same sensor and region, constructing a classification scene of the same sensor at different times. The detailed categories and sample size distribution are shown in Table II. The pseudocolor image and GT map are shown in Fig. 7.

3) *Geographical Divergence Scenario (YC-HHK Dataset):* This dataset simulates the most challenging scene of geographical divergence. The data are from the same sensor but were collected from two distinct geographical wetland scenes: Yancheng and Huanghekou. This reflects significant distribution differences caused by varying geographical environments, ecosystems, and imaging conditions under the same sensor, establishing a highly complex cross-scene classification task of the same sensor, different geography. The detailed categories and sample size distribution are shown in Table II. The pseudocolor image and GT map are shown in Fig. 8.

These three scenes not only meet the practical requirements of multisource, multitemporal, and multiregional wetland monitoring, but also provide a rigorous experimental basis for comprehensively evaluating the adaptability of DG methods in real complex scenes.



(a) (b) (c)

Fig. 9. Performance comparison of MixStyle module parameters. (a) YC. (b) HHK. (c) YC-HHK.

B. Parameter Tuning

The network's performance is influenced by its structural design. This section assesses model performance through the adjustment of hyperparameters, including input patch size, selection range of adaptive thresholds, and window selection size of spatial neighborhoods. We employ overall accuracy (OA), average accuracy (AA), and Kappa coefficient (K) as evaluation metrics, with higher values signifying superior classification performance.

1) MixStyle Module Parameter Comparison: The MixStyle module enhances the adaptability of the model to spectral and textural variations in unknown scenes by implicitly augmenting the style diversity of training data. This module comprises two key hyperparameters: the mixing probability p and the distribution coefficient α. Specifically, p controls whether to perform style mixing for the current batch, while α affects the mixing strength by sampling the mixing coefficient from a Beta distribution. To investigate its impact on model performance, we systematically conducted parameter ablation experiments. We set p to vary from 0.1 to 1.0 with a step size of 0.1, and α to take values of {0.01, 0.1, 1, 10, 100}, using OA as the evaluation metric. The experimental results are shown in Fig. 9. It can be observed that on the YC dataset, the optimal parameter combination is $p = 0.5$ and $\alpha = 10$. However, on the other two datasets, the optimal parameter combination is $p = 0.7$ and $\alpha = 100$. This may be because the style distribution differences across geographical scenes are more complex, requiring stronger style mixing to simulate them. This difference can be explained by the nature and complexity of the domain shifts. For

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TABLE III
OA (%) OF DIFFERENT WEIGHTS BETWEEN MULTIPLE LOSS FUNCTIONS

Scene	Parameter λ				
	0.01	0.1	1	10	100
YC	83.71±2.57	82.56±1.46	**85.67±0.90**	83.51±2.35	84.70±1.04
HHK	90.44±1.30	88.99±1.12	**91.31±0.92**	90.69±1.09	86.51±1.16
YC-HHK	80.81±1.39	81.68±0.86	**82.01±2.01**	81.40±0.44	80.76±0.93

TABLE IV
OA (%) OF DIFFERENT INPUT PATCH SIZES

Scene	patch size				
	7	9	11	13	15
YC	85.11±1.05	85.33±0.56	83.75±1.06	**85.67±0.90**	85.58±1.34
HHK	87.55±1.45	86.20±0.75	83.85±1.88	**91.31±0.92**	91.20±0.65
YC-HHK	81.21±1.03	79.30±1.55	80.84±0.63	**82.01±2.01**	81.09±1.08

TABLE V
OA (%) OF DIFFERENT β CONFIGURATIONS

β weights				OA (%)		
β_{conc}	β_{rule}	β_{inst}	β_{caus}	YC	HHK	YC-HHK
0.25	0.25	0.25	0.25	84.34±1.32	90.23±1.28	79.01±1.56
0.10	0.30	0.20	0.40	83.12±1.67	89.12±1.56	77.34±2.05
0.20	0.30	0.40	0.10	84.34±1.32	90.23±1.28	79.01±1.56
0.30	0.20	0.10	0.40	83.78±1.45	89.87±1.42	78.12±1.89
0.40	0.30	0.20	0.10	85.67±0.90	91.31±0.92	82.01±2.01

sensor discrepancy, the style variations are primarily caused by different imaging instruments but remain within the same geographical region, resulting in relatively bounded and structured distribution shifts. A moderate mixing probability and intensity are sufficient to simulate such limited variations. In contrast, temporal variation involves seasonal and phenological changes that alter both spectral and textural characteristics more dynamically, while geographical divergence introduces entirely different landscape contexts. These complex shifts produce a wider and more irregular distribution of feature styles. Therefore, a higher mixing probability and a larger are required to enhance the model's coverage over unseen style distributions.

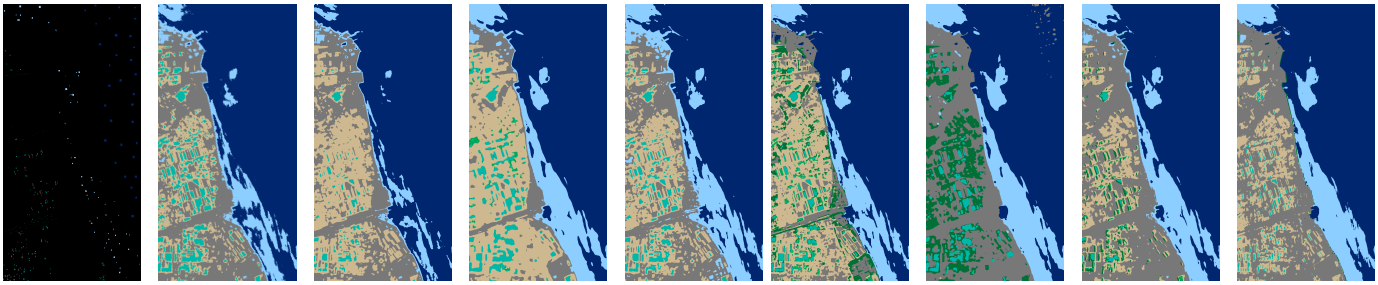
Based on these findings, we derive a practical hyperparameter selection guideline for applying our method to new scenarios: 1) for low-to-moderate domain shifts, such as sensor discrepancies or imaging condition differences within the same region, we recommend a conservative setting of $p = 0.5$ and $\alpha = 10$, which provides sufficient style augmentation without over-regularizing the feature space; and 2) for complex domain shifts, such as significant temporal variations or cross-geographical transfers, we recommend a stronger setting of $p = 0.7$ and $\alpha = 100$ to better simulate the broader style variations encountered in the target domain. Notably, as demonstrated in Fig. 9, the model performance remains relatively stable across a reasonable range of parameters, indicating that exhaustive tuning is unnecessary. This empirical guideline, derived directly from our comprehensive experiments, enhances the transferability of our method to diverse real-world applications without introducing additional computational overhead.

2) Weights Between Multiple Loss Functions Comparison: As shown in (17), the total loss function is composed of a weighted sum of the source domain classification loss and the hierarchical visual–language alignment loss. Among them, the hyperparameter λ is used to balance the contributions of the two losses to model optimization. To determine the optimal value of λ, we conducted experiments on $\lambda \in \{0.01, 0.1, 1, 10, 100\}$ on three datasets, and the results are shown in Table III. From the results, it can be seen that when the value of λ is too small, the contribution of visual–language alignment loss to the total loss is severely weakened. The model optimization process is mainly dominated by classification loss, which leads to a tendency to learn shallow visual features in the source domain data that are most directly related to classification but may have domain specificity. This allows the model to achieve good fitting results on the source domain, but the learned features lack cross-domain-invariant semantic consistency, resulting in poor performance when generalizing to the target domain. When the value of λ is too large, the optimization process will overly focus on alignment loss, forcing visual features to "hard align" with the semantic space described in the text, resulting in unstable optimization and even performance degradation. Therefore, the setting of $\lambda = 1$ effectively achieves the unity of discriminability and generalization, and is also the key reason why our proposed hierarchical language perception mechanism can significantly improve cross-scene classification performance. This result also indicates that the semantic supervision signal provided by the designed hierarchical text description is of high quality and efficiency, and has equal value as the classification supervision signal.

3) Input Patch Size Comparison: The patch size of input image blocks is one of the important factors affecting the classification performance of HS images. An appropriate size can preserve sufficient spatial contextual information while avoiding introducing too much redundancy or noise. To investigate the impact of different input patch sizes on model classification performance, we tested input patch sizes of 7×7, 9×9, 11×1, 13×13, and 15×15 in YC, HHK, and YC-HHK scenes, respectively. The experimental results are shown in Table IV. It can be seen that a too-small patch size may not contain sufficient spatial structural information, especially for land cover categories with large spatial ranges or complex textures, which limits classification performance. However, an excessively large patch size may introduce too much surrounding information unrelated to the central pixel, increasing the risk of category confusion, especially in the boundary areas of land features. The patch size of 13×13 shows good balance in all three scenes, capturing effective spatial–spectral features while avoiding information redundancy and boundary blurring caused by excessive size, achieving the best classification performance.

4) Weights Between Hierarchical Loss Comparison: To evaluate the robustness of our method with respect to the hierarchical alignment weights, we conducted a sensitivity analysis by varying the weight distribution among the four knowledge levels. Table V reports the OA on three datasets under different β configurations. The conceptual level serves as the fundamental semantic anchor, providing the most direct super-

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Data visualization classification maps for the target scene on the YC Dataset. (a) GT. (b) SDEnet. (c) LLURnet. (d) TSDAnet. (e) ISDGS. (f) ADNet. (g) LDGnet. (h) EHSnet. (i) Proposed.

TABLE VI
OBJECTIVE EVALUATION INDICATORS OF YC DATASET UNDER DIFFERENT ALGORITHMS

Class	Visual-only methods					Visual-Test methods		
	SDEnet	LLURnet	TSDAnet	ISDGS	ADNet	LDGnet	EHSnet	Proposed
1	91.05±10.85	92.88±12.09	76.44±19.27	95.83±6.85	88.33±14.39	93.00±15.34	87.83±13.54	**96.83±6.15**
2	13.19±13.47	4.70±10.50	0.64±1.01	7.00±9.61	**24.05±15.44**	2.39±3.29	3.31±7.41	9.30±10.84
3	1.51±2.98	6.36±38.32	29.69±14.55	11.66±17.81	40.75±35.13	11.21±15.43	23.18±31.93	**50.45±11.71**
4	89.13±7.96	59.90±22.22	44.49±38.32	47.54±21.19	13.55±30.31	90.33±10.37	70.64±27.00	**95.14±1.70**
5	38.03±12.29	29.14±31.25	**75.38±21.54**	54.44±21.26	63.24±25.36	16.06±28.94	34.61±29.83	17.26±32.34
6	99.48±1.15	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.33±5.68	**100.00±0.00**	**100.00±0.00**
7	82.06±7.02	73.51±9.25	**86.80±5.71**	72.76±6.88	80.09±5.18	82.14±8.68	74.88±12.61	78.80±7.03
OA(%)	84.79±3.08	78.54±1.86	80.18±4.95	77.43±4.90	75.34±5.32	83.05±1.62	80.27±2.80	**85.67±0.90**
AA(%)	59.21±4.85	55.21±9.75	59.06±5.09	55.46±7.03	58.57±7.09	56.01±3.62	56.35±4.12	**71.27±6.51**
K(×100)	78.63±4.36	69.83±2.76	72.63±6.63	68.43±6.85	66.07±7.17	76.30±2.07	72.23±3.81	**79.76±1.32**

vision signal aligned with the classification objective. The rule level captures domain-invariant discriminative attributes that are critical for generalization. These two levels establish the core semantic foundation, hence receiving higher weights. The instance level, while beneficial for enriching appearance diversity, is more susceptible to domain-specific variations, and thus receives a moderate weight. The causal level provides the deepest mechanistic understanding, but its supervision is most abstract and relies on the grounding provided by the other levels, hence receiving the lowest weight to avoid over-regularization. Therefore, for the hierarchical weights β, we set $[\beta_{\text{conc}}, \beta_{\text{rule}}, \beta_{\text{inst}}, \beta_{\text{caus}}] = [0.4, 0.3, 0.2, 0.1]$.

C. Comparison of Classification Performance

To comprehensively evaluate the proposed KHLDGNet, we conducted a systematic comparison with various SOTA DG methods. The competing methods are categorized into two groups for clarity: 1) visual-only DG methods, including SDEnet [27], LLURnet [26], TSDAnet [28], ISDGS [30], and ADNet [31], which operate solely on visual inputs; and 2) vision–language DG methods, including LDGnet [34] and EHSnet [35], which also integrate textual information. This comparative setup allows us to isolate and assess the specific benefits conferred by our novel hierarchical knowledge-guided fusion paradigm against both traditional visual DG baselines and existing multimodal approaches.

1) Analysis of Experimental Results in YC Dataset: YC dataset is characterized by domain shift from different sensors. The comparison of objective evaluation indicators of different algorithms is shown in Table VI, and its visual comparison is shown in Fig. 12. KHLDGNet achieves the best overall performance, surpassing all compared methods. Notably, it demonstrates significant improvements in challenging categories such as Reed and Fallow land. While visual-only methods like TSDAnet and ADNet show competitive OA, their AA is considerably lower, indicating poorer per-class balance. Existing multimodal methods show only marginal gains over visual-only methods in this scene. The superior performance of KHLDGNet, particularly its higher AA, underscores the advantage of its deep hierarchical semantic fusion. By aligning visual features with rule-based and causal descriptions, the model focuses on essential, sensor-invariant attributes of geographic entities rather than superficial spectral patterns, leading to more robust and balanced classification under sensor-induced spectral distortion.

2) Analysis of Experimental Results in HHK Dataset: HHK dataset is characterized by domain shift from temporal variation. The comparison of objective evaluation indicators of different algorithms is shown in Table VII, and its visual comparison is shown in Fig. 11. The advantage is most pronounced in categories highly sensitive to temporal dynamics, such as salt flat evaporation pond and tide ditch, where KHLDGNet outperforms the second-best method by large margins. Several visual-only methods, such as SDEnet and LLURnet, fail completely with 0% accuracy in categories like salt flat filtration pond, highlighting their vulnerability to significant appearance changes over time. While EHSnet shows strong results in some categories, its overall AA lags. KHLDGNet's robustness stems from its causal-level textual guidance, which encodes the underlying physical or functional

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TABLE VII

OBJECTIVE EVALUATION INDICATORS OF HHK DATASET UNDER DIFFERENT ALGORITHMS

Class	Visual-only methods					Visual-Test methods		
	SDEnet	LLURnet	TSDAnet	ISDGS	ADNet	LDGnet	EHSnet	Proposed
1	97.63±1.99	98.73±1.41	**99.81±0.26**	97.19±4.54	97.84±4.45	79.62±15.66	90.23±13.31	93.39±5.40
2	0.78±1.74	0.00±0.00	1.25±2.47	0.11±0.26	4.49±6.39	12.34±16.76	**19.84±14.30**	12.34±5.91
3	6.23±8.92	3.84±6.83	1.23±1.68	3.07±5.64	12.23±14.64	46.92±23.19	41.30±33.52	**62.23±13.12**
4	4.41±7.83	1.84±4.12	0.00±0.00	0.09±0.21	16.40±25.95	**93.06±8.87**	64.41±45.94	89.32±6.03
5	25.93±30.20	0.76±1.45	0.95±1.88	2.92±4.67	14.59±11.08	39.52±11.70	23.97±14.72	**40.43±1.82**
6	97.17±3.50	95.81±4.35	99.65±0.43	**99.87±0.28**	80.00±44.72	73.13±18.54	69.43±41.23	92.62±6.01
7	94.56±5.37	94.13±2.52	85.76±4.32	89.15±5.04	63.90±14.47	84.22±17.82	83.12±14.00	**98.62±1.40**
8	5.88±13.13	7.98±7.30	9.67±18.25	0.16±0.36	0.08±0.17	53.22±32.25	23.46±24.75	**58.46±13.67**
OA(%)	82.47±3.97	82.51±1.33	76.74±2.95	79.09±4.23	60.44±8.01	78.22±13.45	76.30±10.22	**91.31±0.92**
AA(%)	41.36±6.24	37.89±1.52	37.29±2.32	36.57±1.91	36.19±7.54	60.25±7.38	51.89±14.62	**72.95±7.75**
K(×100)	62.54±6.28	62.45±1.66	54.82±3.72	57.66±5.82	37.24±6.45	60.81±17.09	57.03±14.94	**80.81±1.82**

TABLE VIII

OBJECTIVE EVALUATION INDICATORS OF YC-HHK DATASET UNDER DIFFERENT ALGORITHMS

Class	Visual-only methods					Visual-Test methods		
	SDEnet	LLURnet	TSDAnet	ISDGS	ADNet	LDGnet	EHSnet	Proposed
1	0.00±0.00	5.88±8.26	0.00±0.00	0.00±0.00	0.00±0.00	71.55±23.91	24.67±14.43	**77.94±26.37**
2	**94.44±8.85**	20.58±7.32	82.32±25.68	93.14±6.25	57.08±39.96	55.98±24.36	60.31±32.71	47.08±15.29
3	**12.89±28.84**	0.34±0.39	0.00±0.00	0.00±0.00	7.80±17.44	6.14±6.61	4.31±4.81	7.41±6.61
4	10.32±17.46	10.96±19.20	2.58±4.04	11.29±11.85	4.84±9.12	**53.55±32.90**	42.26±30.30	38.38±25.88
5	92.78±9.89	82.33±19.29	91.49±12.74	64.33±21.95	78.06±20.20	99.61±0.41	90.89±8.60	**99.64±0.42**
6	7.48±10.26	12.00±16.63	10.31±14.47	32.28±14.56	21.85±20.48	31.35±22.69	24.82±25.17	**54.21±8.95**
OA(%)	70.64±4.29	58.41±10.31	68.27±7.16	53.76±12.57	59.44±12.09	79.69±1.82	70.42±4.95	**82.01±2.01**
AA(%)	36.32±4.71	22.02±4.10	31.11±4.12	33.51±2.16	28.27±6.65	53.03±7.50	41.24±5.86	**59.64±13.71**
K(×100)	39.20±2.68	22.90±6.64	37.02±5.49	25.34±9.08	28.44±9.95	59.56±4.56	44.31±6.57	**64.51±4.10**

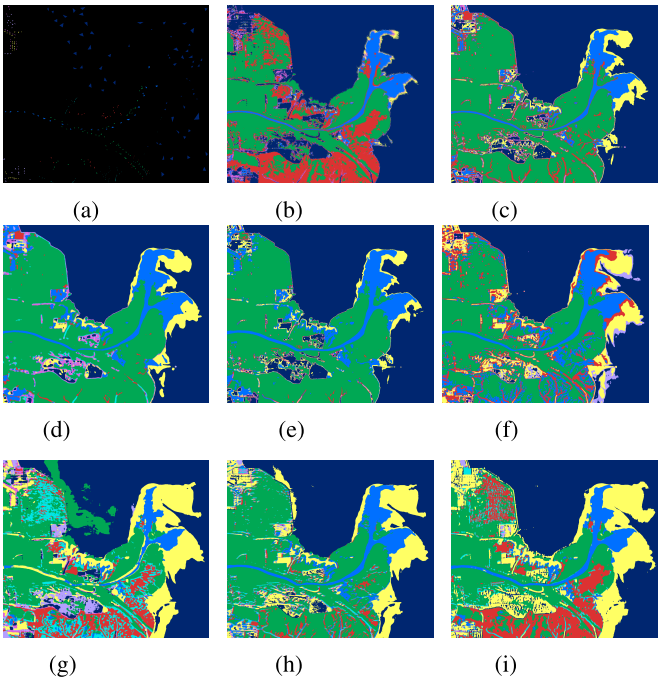


Fig. 11. Data visualization classification maps for the target scene on the HHK dataset. (a) GT. (b) SDEnet. (c) LLURnet. (d) TSDAnet. (e) ISDGS. (f) ADNet. (g) LDGnet. (h) EHSnet. (i) Proposed.

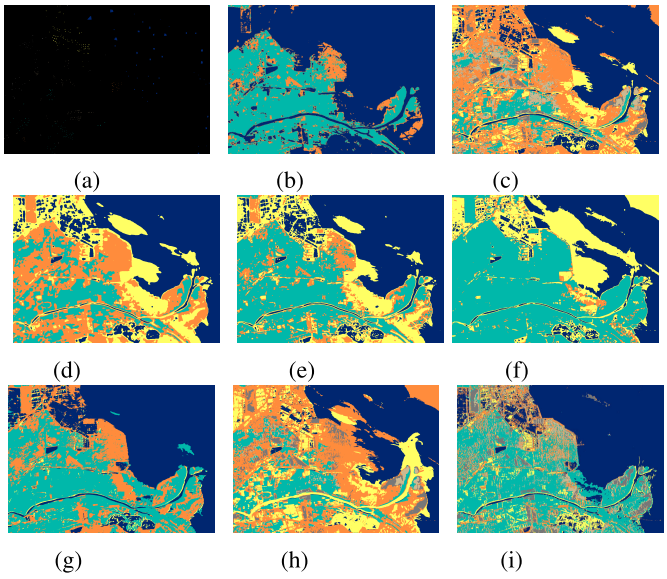


Fig. 12. Data visualization classification maps for the target scene on the YC-HHK dataset. (a) GT. (b) SDEnet. (c) LLURnet. (d) TSDAnet. (e) ISDGS. (f) ADNet. (g) LDGnet. (h) EHSnet. (i) Proposed.

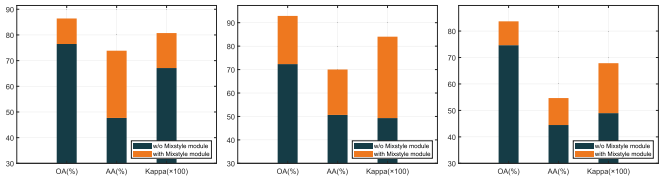
mechanisms that remain consistent across seasons, such as evaporation processes and tidal scouring. This allows the model to maintain semantic focus despite substantial changes in low-level visual appearance, demonstrating the critical role of deep knowledge fusion in handling temporal variation.

3) Analysis of Experimental Results in YC-HHK Dataset: YC-HHK dataset is characterized by a domain shift from a different geographical region. The comparison of objective evaluation indicators of different algorithms is shown in Table VIII, and its visual comparison is shown in Fig. 14. This scene accentuates the limitations of methods that lack structured semantic grounding. Visual-only methods, which rely on learning statistical invariances from visual data, struggle

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TABLE IX
COMPARISON OF MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY

	SDEnet	LLURnet	TSDAnet	ISDGS	ADNet	LDGnet	EHSnet	Proposed
Params(k)	3143.68	494.28	2046.98	494.28	490.89	2301.95	2002.94	2543.62
MFLOPs	157.11	12.54	194.06	2.54	11.24	239.19	379.97	328.77
Training Time(s/epoch)	3.50	2.16	3.72	0.95	2.48	11.41	8.18	15.75
Inference Time(s)	1.96	0.06	0.11	0.04	0.10	1.23	2.13	1.16



(a) (b) (c)

Fig. 13. Influence of with/without MixStyle feature generalization module on classification accuracies. (a) YC. (b) HHK. (c) YC-HHK.

TABLE X
OBJECTIVE EVALUATION INDICATORS ON THE HOUSTON13–18 CROSS-DOMAIN BENCHMARK

Method	OA (%)	AA (%)	K (×100)
SDEnet	73.95±4.15	70.77±5.53	58.55±4.60
LLURnet	69.87±1.87	67.53±3.27	53.01±2.01
TSDAnet	66.45±0.52	42.59±8.73	42.13±5.45
ISDGS	65.72±3.87	42.85±15.12	39.76±9.56
ADNet	60.43±1.95	63.28±2.05	43.07±1.06
LDGNet	77.22±1.67	65.85±2.47	61.02±3.46
EHSnet	63.76±2.50	65.80±0.24	48.46±2.42
Proposed	**78.69±0.98**	**67.34±3.89**	**63.02±2.68**

with the large interclass appearance variance across different geographies. While vision–language methods LDGnet and EHSnet incorporate text, their use of relatively shallow semantic prompts provides insufficient guidance for such complex shifts, such as category names or simple attributes. In contrast, KHLDGNet's hierarchical knowledge system provides a multifaceted semantic anchor. The rule-level descriptions enforce stable spatial–spectral characteristics, while the causal-level explanations bind these characteristics to universal geographical or ecological principles, enabling the model to generalize across fundamentally different wetland landscapes.

We also noticed that some purely visual methods, such as LLURnet and SDEnet, have an accuracy of 0% in certain categories. The phenomenon of zero accuracy is mainly due to two factors: 1) the sample size of certain categories in the target domain is extremely limited, e.g., the "salt beach filter" category in HHK only has five target samples; and 2) severe domain shift only affects visual methods lacking semantic prior, resulting in complete misclassification under large spectral texture changes. In contrast, our KHLDGNet maintains robust performance even under challenging conditions guided by hierarchical semantic knowledge.

4) Analysis of Computing Efficiency: In order to comprehensively evaluate the feasibility of KHLDGNet in practical applications, this section compares and analyzes KHLDGNet with existing DG methods from two dimensions: training cost and inference efficiency. We compared the computational cost of all methods on the YC dataset, including Params, MFLOPs, training time per epoch, and inference time on the full test set. Table IX presents the results, and it can be seen that KHLDGNet has 2.54-M parameters and 328.77 MFLOPs, which is moderately higher than lightweight methods but comparable to other vision–language models. The training time per epoch is the highest among all methods, mainly due to the additional hierarchical contrastive alignment across four text levels. However, training is performed offline only once, and the increased cost is acceptable given the substantial gains in generalization and interpretability. More importantly, during inference, the text encoder and alignment module are completely removed, resulting in a test time of only 1.16 s, faster than SDEnet, LDGnet, and EHSnet, and sufficiently efficient for practical wetland monitoring applications. Thus, KHLDGNet achieves a favorable tradeoff between accuracy and computational cost without sacrificing deployment efficiency.

5) Analysis of Experimental Results on Standard Dataset: To evaluate the broader applicability of KHLDGNet beyond wetland scenarios, we additionally conducted experiments on the Houston13–18 cross-domain benchmark. Houston2013 and Houston2018 are two widely used hyperspectral datasets covering the same geographical area but acquired at different times with different sensors, presenting a realistic domain shift scenario. Following the common practice in cross-scene HSI classification, we use the seven shared classes between the two datasets for evaluation. For all comparison algorithms and our proposed method, we used a patch size of 13, a learning rate of $1e^{-3}$, and an epoch of 30. Table X reports the OA, AA, and Kappa of our KHLDGNet compared with representative visual-only and vision–language DG methods. As shown, KHLDGNet achieves the highest results. This confirms that the hierarchical knowledge guidance is not limited to wetland-specific semantics but can generalize to urban land-cover classification as well. The consistent improvement over existing methods validates the general effectiveness of our hierarchical vision–language alignment framework. These results demonstrate that KHLDGNet possesses broad applicability across different HSI classification tasks and domain shift scenarios.

D. Ablation Study and Explainability Analysis

1) Effect of Core Modules: To isolate the contribution of each core component, we compared four configurations: 1) baseline (w/o MixStyle, w/o Text); 2) +Text (w/o MixStyle, w/ Text); 3) +MixStyle (w/ MixStyle, w/o Text); and 4) full

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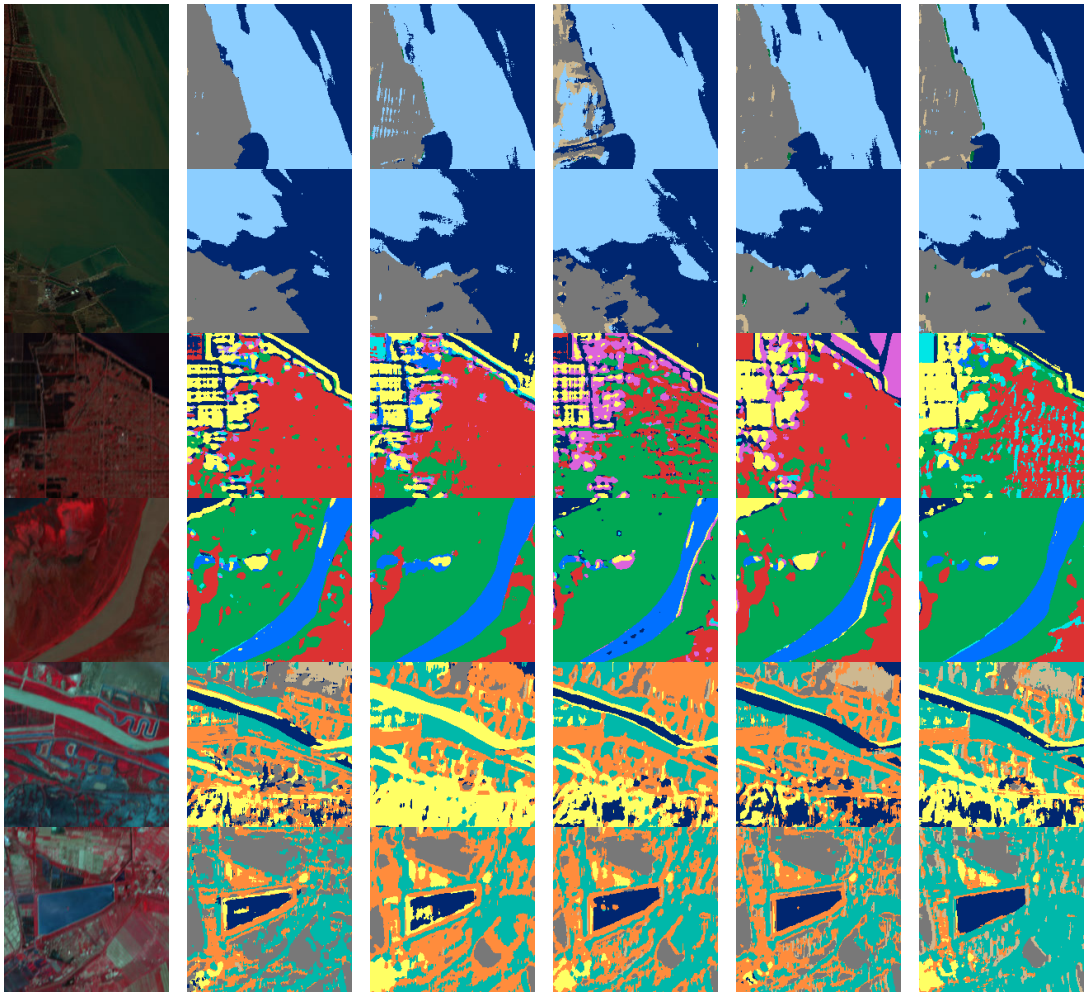


Fig. 14. Visualization results of typical scenes under different texts. (a) HS pseudocolor image. (b) w/o conceptual-level text. (c) w/o rule-level text. (d) w/o instance-level text. (e) w/o causal-level text. (f) With all texts.

TABLE XI
ABLATION STUDY ON CORE MODULES

	YC			HHK			YC-HHK		
	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
w/o MixStyle & w/o Text	69.04±4.78	49.40±9.89	56.11±7.35	59.42±18.23	50.55±5.02	39.56±15.82	59.66±9.66	48.46±5.35	42.30±7.64
w/o MixStyle only	77.49±8.61	54.31±7.49	67.33±13.23	77.55±1.76	55.24±2.21	48.33±4.38	69.06±5.58	38.85±6.54	42.21±9.16
w/o Text only	77.56±8.74	52.10±3.06	63.74±13.11	66.17±13.21	44.75±14.96	45.59±14.58	76.70±2.36	46.62±2.41	52.92±4.44
with all	**85.67±0.90**	**71.27±6.51**	**79.76±1.32**	**91.31±0.92**	**72.95±7.75**	**80.81±1.82**	**82.01±2.01**	**59.64±13.71**	**64.51±4.10**

KHLDGNet (w/ both) Table XI reports the OA, AA, and KAPPA for each configuration.

As shown in Table XI, the baseline yields the lowest performance across all three datasets, with particularly poor generalization on HHK and YC-HHK, highlighting the severe domain shift challenges. Adding hierarchical text alignment alone brings substantial gains. This confirms that the proposed four-level knowledge hierarchy provides effective domain-invariant semantic guidance. The full KHLDGNet further improves, demonstrating that MixStyle contributes complementary robustness against spectral-textural variations across different domain shift types. Notably, the largest absolute gains from the full model occur on HHK, where style mixing is most beneficial. These results clearly validate the independent and synergistic contributions of both core modules.

2) Effect of Multimodal MixStyle Feature Generalization Module: The MixStyle module enhances the style diversity of training data by implicitly mixing styles at the feature layer, allowing the visual encoder to learn more robust representations of interdomain spectral and texture changes. By simulating the feature distribution changes under different sensors, seasons, and lighting conditions, the module significantly alleviates the problem of domain offset, thus greatly improving the generalization performance of the model on unknown scenes. To verify the effectiveness of the proposed MixStyle module in cross-scene HS wetland classification, we conducted

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TABLE XII
COMPARISON OF MODEL PERFORMANCE UNDER DIFFERENT TEXT COMBINATIONS

Text Combinations				YC			HHK			YC-HHK		
Conceptual	Rule	Instance	Causal	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
✓				84.16±2.04	54.20±6.72	77.40±3.01	88.36±1.74	62.12±3.35	73.75±3.76	76.38±2.59	46.60±9.05	52.89±6.62
	✓			83.46±2.03	63.73±9.37	76.68±2.73	85.96±3.46	49.27±12.32	68.48±8.51	78.56±1.17	47.79±6.84	57.64±3.17
		✓		82.34±1.10	57.54±1.60	75.22±1.44	86.86±0.88	56.44±10.57	69.56±2.06	63.60±12.48	50.38±5.40	44.83±11.20
			✓	83.05±2.95	55.21±4.03	76.07±4.23	89.37±1.85	62.22±7.34	76.18±4.57	75.60±4.19	49.03±6.94	53.58±6.70
✓	✓			83.25±1.98	56.14±4.83	76.35±2.69	90.54±1.46	67.69±2.23	79.42±3.03	75.68±3.84	49.82±2.86	54.74±3.81
✓		✓		84.17±3.36	55.44±6.29	77.60±4.55	90.00±0.95	66.44±3.20	77.61±2.66	77.92±1.65	51.78±7.10	57.09±3.78
✓			✓	84.24±2.15	60.01±6.76	77.79±3.16	90.97±0.86	69.76±1.63	80.49±1.79	79.41±1.51	51.34±6.41	59.03±3.60
	✓	✓		85.66±1.30	57.83±2.15	79.68±2.04	90.81±0.65	67.97±2.58	79.73±1.44	80.92±0.88	54.89±3.18	62.25±2.47
	✓		✓	85.58±1.34	60.03±4.73	79.59±2.07	90.16±0.56	68.22±1.93	78.50±1.17	78.43±2.45	49.84±5.00	57.85±3.81
		✓	✓	84.73±2.29	67.60±7.95	77.88±2.99	90.45±1.01	67.64±2.88	78.85±2.50	79.67±1.68	50.30±4.27	58.76±3.47
	✓	✓	✓	85.13±1.31	70.49±8.21	79.01±1.82	89.31±4.13	65.25±3.13	79.12±2.49	80.57±1.02	53.85±4.26	61.94±1.31
✓		✓	✓	84.55±2.01	64.57±9.07	78.30±2.75	91.45±0.52	68.46±1.88	81.19±1.30	81.98±0.64	55.62±4.38	63.81±1.26
✓	✓		✓	85.13±2.82	61.56±12.86	78.88±3.90	91.75±0.19	68.88±2.41	81.49±0.80	81.88±1.48	57.26±5.51	64.44±2.89
✓	✓	✓		84.67±2.19	70.89±7.34	78.34±3.08	91.80±1.01	68.41±3.22	81.67±2.43	80.86±0.69	53.79±4.78	61.47±1.59
✓	✓	✓	✓	**85.67±0.90**	**71.27±6.51**	**79.76±1.32**	**91.31±0.92**	**72.95±7.75**	**80.81±1.82**	**82.01±2.01**	**59.64±13.71**	**64.51±4.10**

ablation experiments in three scenes. The experiment compared the classification performance of the model with and without the MixStyle module, and the experimental results are shown in Fig. 13. It can be seen that the introduction of the MixStyle module significantly improves the classification performance in all three scenes, indicating that the MixStyle module can effectively enhance the model's adaptability to cross-scene style changes.

From the perspective of explainability, the introduction of the MixStyle module not only improves the model's generalization ability, but also enhances the semantic consistency of its decisions. This module perturbs the style statistics in shallow features, forcing the model to rely more on high-level semantic features for classification, rather than being limited to overfitting specific styles in the source domain. The MixStyle module helps the model to detach the apparent associations related to the domain, making it more focused on the essential semantic features shared across scenes, thereby improving performance and enhancing the traceability and interpretability of model decisions.

3) Effect of Hierarchical Text Knowledge: In order to further verify the role of hierarchical text knowledge in cross-scene HS wetland classification, this section designed five experimental settings: removing concept-level, rule-level, instance-level, and causal-level texts, respectively, and comparing them with the complete four-layer text setting. Table XII shows the impact of different text hierarchy configurations on model performance in three scenes.

To comprehensively assess both the individual and joint contributions of the four knowledge levels, we conducted a systematic combination ablation study with 15 configurations, ranging from single-level to full four-level combinations. The combination ablation study unequivocally validates our hierarchical design. Each of the four levels contributes unique and complementary semantic guidance; their combination yields synergistic gains, and the full hierarchy is indispensable for achieving optimal generalization across diverse domain shift scenarios. Single-level configurations reveal that the conceptual level alone achieves the highest OA among single-level variants, while the causal level alone performs poorly on YC-HHK, confirming that causal descriptions, while providing deep semantic insights, are meaningful only when grounded in foundational conceptual and rule-level knowledge. The full four-level hierarchy achieves the highest OA on all three datasets, confirming that all four levels provide complementary and nonredundant semantic guidance. Level importance varies with domain shift type. For temporal variation, the instance and causal levels are particularly important because seasonal and phenological changes dramatically alter visual appearance, requiring both rich visual exemplars and mechanistic explanations to maintain semantic focus. For geographical divergence, the rule and causal levels play a more critical role, as cross-regional differences primarily affect the spatial–spectral patterns defined by rule-level attributes, while causal mechanisms remain geographically invariant. For sensor discrepancy, all levels contribute more evenly, as the domain shift is relatively bounded. This observation further justifies the necessity of a multilevel hierarchy: different domain shifts affect different levels of semantic knowledge, and a flat semantic representation would be insufficient to address all shift types simultaneously.

At the same time, in order to verify the influence of different texts on typical terrain in different scenes, we visualized some typical scenes, and the results are shown in Fig. 14. It can be seen that the importance of different text layers varies in different scenes: the concept layer is more critical in cross-geographical scenes, the rule layer plays a significant role in temporally changing scenes, the instance layer may introduce specific biases in certain situations, and the causal layer provides stable cross-DG ability. This reflects the complexity of real-world geographic scenes, which require multilevel and multiangle semantic guidance to cope with various domain shifts.

To rigorously evaluate the interpretability of our method, we conducted an extended similarity analysis. For the YC dataset, we randomly selected 20 correctly classified pixels per category across all seven categories. For each pixel, we computed the cosine similarity between its visual feature and the text embeddings of its ground-truth class at each of the four knowledge levels. Table XIII reports the mean

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TABLE XIII
COSINE SIMILARITY BETWEEN VISUAL FEATURES AND HIERARCHICAL TEXT EMBEDDINGS ACROSS DIFFERENT CATEGORIES IN YC DATASET

Category	Knowledge Level			
	Conceptual	Rule	Instance	Causal
Architecture	**0.92±0.03**	0.71±0.05	0.61±0.07	0.88±0.04
River	0.86±0.05	0.69±0.06	0.58±0.09	**0.89±0.04**
Reed	**0.91±0.03**	0.72±0.05	0.65±0.07	0.89±0.04
Paddy	0.89±0.04	0.74±0.06	0.63±0.08	**0.91±0.03**
Fallow land	**0.91±0.05**	0.70±0.07	0.59±0.09	0.90±0.04
Sea	0.85±0.06	0.66±0.07	0.55±0.10	**0.88±0.05**
Offshore water	0.85±0.06	0.67±0.07	0.56±0.09	**0.86±0.05**
Overall Average	0.87±0.05	0.69±0.06	0.60±0.08	**0.88±0.04**

and standard deviation of these similarities for each category and level. For the vast majority of categories, the causal level yields the highest average similarity, followed by the conceptual level, while the instance-level consistently exhibits the lowest similarity. This indicates that the model does not rely on superficial visual details or specific exemplars for decision-making; instead, it grounds its predictions on deep, domain-invariant knowledge, such as the ecological function of vegetation types, the hydrological formation mechanisms of tidal landforms, or the planning principles underlying built structures. Meanwhile, we note some category-specific variations. For categories such as Architecture, the conceptual level occasionally exhibits slightly higher similarity than the causal level. This is reasonable, as these anthropogenic categories are less defined by natural formation mechanisms and more by their functional identity, making the conceptual-level description the most direct semantic anchor. This extended statistical analysis provides robust quantitative evidence that KHLDGNet's decisions are traceable to specific semantic descriptions, particularly the causal and conceptual levels, confirming the interpretability claim of our method.

IV. CONCLUSION

This article has proposed a KHLDGNet to tackle the dual challenges of domain shift and limited interpretability in cross-scene hyperspectral wetland classification. A style-enhanced visual encoder is designed, which improves robustness against spectral and textural variations through implicit feature-level style mixing. For the first time in hyperspectral wetland classification, a structured four-level geographic knowledge system is introduced that encompasses conceptual, rule, instance, and causal knowledge. This hierarchical semantic guidance enables the model to progressively reason from shallow visual patterns to the formation mechanisms of geographic entities, thereby enhancing adaptability to cross-scene changes. By leveraging textual prior knowledge to guide visual feature learning, an intrinsically interpretable vision–language alignment mechanism is achieved, making classification decisions traceable to specific semantic descriptions and opening a new pathway for XGeoAI. Experimental results on three real-world wetland datasets demonstrate that KHLDGNet significantly outperforms existing DG methods, not only in classification accuracy but also in generalization capability and interpretability.

Nevertheless, the current textual knowledge system is primarily tailored to wetland scenes, and the construction of textual descriptions still relies on expert knowledge. Future work will extend the hierarchical semantic guidance framework to other RS tasks, explore automated or semi-automated methods for semantic description generation, and integrate large language models (LLMs) to enhance semantic modeling. These efforts aim to advance the development of interpretable, generalizable, and trustworthy geographic intelligence systems.

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